DE–8334 **11**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that HK is a subgroup of G if and only if $HK = KH$.

   (b) Prove that N of G is a normal subgroup of G if and only if every left coset of N in G is a right coset of N in G.

2. (a) Prove that the number of conjugate classes in S_n is $P(n)$, the number of partitions of n.

   (b) Suppose that G is the internal direct product of $N_1, \ldots N_n$. Then prove for $i \neq j$, $N_i \cap N_j = (e)$ and if $a \in N_i, b \in N_j$ then $ab = ba$.

3. (a) Prove that a finite integral domain is a field.

   (b) If R is a commutative ring with unit element and M is an ideal of R, then prove that M is a maximal ideal of R if and only if R / M is a field.

4. (a) Let R be a Euclidean ring and $a, b \in R$. If $b \neq 0$ is not a unit in R, then prove that $d(a) < d(b)$.

   (b) If R is a unique factorization domain, then prove that $R[x]$ is a unique factorization domain.

5. (a) If $v_1, v_2, \ldots v_n$ are in V, then prove that either they are linearly independent or some v_k is a linear combination of proceeding ones $v_1, \ldots . v_{k-1}$.

(b) If $T \in A(V)$ has all its characteristic roots in F, then prove that there is a basis of V in which the matrix of T is triangular.

6. (a) If $T \in A(V)$ is such that $(vT, v) = 0$ for all $v \in V$ then prove that $T = 0$.

(b) Prove that $T \in A(V)$ is unitary if and only if $TT^* = 1$ and $T \in A(V)$ is Hermitian then prove all its characteristic roots are real.

7. (a) Prove that a polynomial of degree n over a field can have atmost n roots in any extension field.

(b) If $P(x) \in F[x]$, then and if K is an extension of F then prove that for any element $b \in K$, $P(x) = (x - b)\, q(x) + P(b)$ where $q(x) \in K[x]$ and where $\deg q(x) = \deg P(x) - 1$.

8. (a) Prove that the fixed field of G is a subfield of K.

(b) Prove that K is a normal extension of F if and only if K is the splitting field of some polynomial over F.

DE–8335 **12**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

REAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Prove that a set E is open if and only if its complement is closed. (10)

   (b) Prove that for any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open and for any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed. (10)

2. (a) State and prove Heine-Borel theorem. (10)

   (b) Suppose f is a continuous mapping of a compact metric space X into a metric space Y. Then prove that $f(X)$ is compact. (10)

3. (a) State and prove chain rule of differentiation. (5)

   (b) State and prove Rolle's theorem. (5)

   (c) State and prove Taylor's theorem. (10)

4. State and prove inverse function theorem. (20)

5. (a) Prove that $f \in \mathcal{R}(\alpha)$ on $[a, b]$ if and only if for every $\in > 0$ there exists a partition P such that $U(P, f, \alpha) - L(P, f, \alpha) < \in$. (10)

(b) If $f \in \mathbf{R}(\alpha)$ and $g \in \mathbf{R}(\alpha)$ on $[a, b]$, then prove the following :

(i) $f\,g \in \mathcal{R}(\alpha)$

(ii) $|f| \in \mathcal{R}(\alpha)$ and $\left|\int_a^b f\,d\alpha\right| \leq \int_a^b |f|\,d\alpha$. (10)

6. Prove that there exists a real continuous function on the real line which is nowhere differentiable. (20)

7. (a) Prove that the outer measure of an interval is its length. (10)

(b) If E_1 and E_2 are measurable then prove that $E_1 \cup E_2$ is measurable. (10)

8. (a) State and prove bounded convergence theorem. (10)

(b) State and prove Fatou's Lemma. (10)

DE–8336 **13**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

DIFFERENTIAL EQUATIONS AND NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Find the solution of the following initial value problems :

   (i) $y'' + (4i + 1)y' + y = 0, y(0) = 0, y'(0) = 0$

   (ii) $y'' + 10y = 0, y(0) = \pi,\ y''(0) = \pi^2$. (10)

   (b) Derive Legendre equation, with an example. (10)

2. (a) Explain Bessel equation, with an example. (10)

   (b) Prove : The equation $y + \left(p + \frac{1}{2} - \frac{x^2}{4}\right)y = 0$, where p is a constant, certainly has a series solution in the form $y = \Sigma_j\, a_j x_j$, Prove that the coefficients a_j are related by the three-term recursion formula $(n+1)(n+2)\,a_{n+2} + \left[p + \frac{1}{2}\right]a_n - \frac{1}{4}\,a_{n-2} = 0$. (10)

3. (a) Explain Jacobi's method, in details with an example. (10)

   (b) Solve $p^2x + q^2y = z$ using Jacobi's method. (10)

4. Obtain the D' Alembert's solution of the one dimensional wave equation and interpret it physically. (20)

5. Find a root of the equation $e^x - 3x^2 = 0$, which is approximately, correct three places of decimal by using
   (a) Newton's method (10)
   (b) Fixed point method. (10)

6. (a) Let $y(x)$ and $z(x)$ be non-trivial solutions of $y'' + q(x)y = 0$ and $z'' + r(x)z = 0$ where $q(x)$ and $r(x)$ are positive functions such that $q(x) > r(x)$. Then $y(x)$ vanishes atleast least once between any two successive zeros $z(x)$. (10)

   (b) Let $u(x)$ be any non-trivial solution of $u'' + q(x)u = 0$ where $q(x) > 0$ for all $x > 0$. If $\int_1^\infty q(x)\,dx = \infty$, then $u(x)$ has infinitely many zeros on the positive x axis. (10)

7. Find the value of $\int_0^2 \frac{dx}{x^2 + 4}$ by dividing the range into 4 equal, parts using
   (a) Simpson's $\frac{1}{3}$ rule (10)
   (b) Trapezoidal rule. (10)

8. Solve the differential equation $\frac{dy}{dx} = \frac{y - x}{y + x}$, $y(0) = 1$, using
   (a) Taylor's method (10)
   (b) Picard's method. (10)

DE–8337 **14**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Use Simplex method to solve the LPP : (10)

Maximize $z = 4x_1 + 10x_2$

Subject to the constraints :

$$2x_1 + x_2 \le 500$$
$$2x_1 + 5x_2 \le 100$$
$$2x_1 + 3x_2 \le 90$$
$$x_1 \ge 0,\ x_2 \ge 0.$$

(b) Explain Dual Simplex Algorithm. (10)

2. (a) Consider the LPP :

Maximize $z = 3x_1 + 5x_2$

Subject to :

$$x_1 \le 4$$
$$3x_1 + 2x_2 \le 18$$
and $x_1, x_2 \ge 0$.

If a new variable x_5 is introduced with $c_5 = 7$ and $a_5 = [1, 2]$, discuss the effect of adding the new variable and obtain the revised solution if any. (10)

(b) Discuss about Integer programming algorithms. (10)

3. (a) What is dynamic programming? Explain the advantages and disadvantages of dynamic programming. (10)

(b) Nation Outdoors School (NOS) is preparing a summer camp-site in the heart of Alaska to train individuals in wilderness survival. NOS estimates that attendance can fall into one of four categories: 200, 250, 300 and 350 persons. The cost of the camp-site will be the smallest when its size meets the demand exactly. Deviations above or below the ideal demand levels incur additional costs resulting from building surplus (unused) capacity or losing income opportunities when the demand is not met. Letting a_1 to a_4 represent the sizes of the camp-sites (200, 250, 300 and 350 persons) and s_1 to s_4 the level of attendance, the following table summarizes the cost matrix (in thousands of dollars) for the situation.

	s_1	s_2	s_3	s_4
a_1	5	10	18	25
a_2	8	7	12	23
a_3	21	18	12	21
a_4	30	22	19	15

Analyzed the problem is using the following four criteria :

(i) Laplace

(ii) Minimax

(iii) Savage

(iv) Harwiez. (10)

4. (a) Solve the following 2×2 game graphically : (10)

$$\begin{array}{c} \\ A_1 \\ A_2 \end{array} \begin{array}{c} \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \begin{bmatrix} 2 & 0 & 0 & -2 \\ 1 & 0 & 0 & 2 \end{bmatrix} \end{array}$$

(b) Derive the linear programming solution of games. (10)

5. (a) Construct the network diagram activities $B, C, \ldots Q$ and N such that the following constraints are satisfied. $B < E, F$; $C < G, L$; $E, G < H$; $L, H < I$, $L < M$; $H < N$; $H < J$; $I, J < P$; $P < Q$. The notation $X < Y$ means that the activity X must be finished before Y can begin. (10)

(b) Explain Critical Path Method (CPM) computations. (10)

6. (a) Consider a single-period model. Show that for the discrete demand the optimal order quantity is determined from $P(D \le y^* - 1) \le \dfrac{p-c}{p+h} \le P(D \le y^*)$. (10)

(b) LubeCar specializes in fast automobile oil change. The garage buys car oil in bulk at \$3 per gallon. A discount price of \$2.50 per gallon is available if LubeCar purchases more than 1000 gallons. The garage services approximately 150 cars per day, and each oil change takes 1.25 gallons. LubeCar stores bulk oil at the cost of \$.02 per gallon per day. Also, the cost of placing an order for bulk oil is \$20. There is a 2-day lead time for delivery. Determine the

optimal inventory policy. The consumption of oil per day is D = 150 cars per day × 1.25 gallons per car = 187.5 gallons per day. We also have

h = \$.02 per gallon per day

k = \$20 per order

L = 2 days

c_1 = \$3 per gallon

c_2 = \$2.50 per gallon

q = 1000 gallons. (10)

7. (a) (i) What do you mean by queuing system? (5)

(ii) Explain: Pure birth process and pure death process. (5)

(b) Discuss about steady-state measures of performance. (10)

8. (a) Explain the queuing model $(M/M/1)$: $(GD/\infty/\infty)$. (10)

(b) Explain the queuing model $(M/M/c)$: $(GD/N/\infty)$. (10)

DE–8338 **15**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

MATHEMATICAL STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) The joint p.d.f of two random variables X and Y is given by $P_r\{x=0, y=1\}=\frac{1}{3}$, $P_r\{x=1, y=-1\}=\frac{1}{3}$ and $P_r\{x=1, y=1\}=\frac{1}{3}$. Find

   (i) Marginal distributions of X and Y.

   (ii) The conditions probability distribution of X given $Y=1$. (10)

   (b) Let the random variables X and Y be independent and their probability density functions are, respectively given by ; $f(x)=\frac{1}{\pi}.\ \frac{1}{\sqrt{1-x^2}}, |x|<1$ and $g(y)=y\exp\left(-\frac{1}{2}y^2\right), \quad y>0$. Find the joint probability density function of Z and W, where $Z=XY$ and $W=X$. Deduce the probability density function of Z. (10)

2. (a) Find the m.g.f. of standard binomial variable $(X-np)/\sqrt{npq}$ and obtain its limiting form as $n \to \infty$. Also interpret the result. (10)

(b) Explain additive property of independent Poisson variables. (10)

3. (a) Prove that $X_i = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } q = 1-p \end{cases}$, then the distribution of the random variable $S_n = X_1 + X_2 + ... + X_n$, where X_i's are independent, is asymptotically normal as $n \to \infty$. (10)

(b) If $X \sim N(\mu, \sigma^2)$, obtain the p.d.f. of the r.v. $U = \frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2$. (10)

4. (a) If X and Y are independent Gamma variates with parameters μ and ν respectively, show that the variables : $U = X + Y$ and $V = \frac{X-Y}{X+Y}$ are independent variables. (10)

(b) Explain Rao-Cramer inequality for a random variable. (10)

5. Let $k > 0$, be a constant and ω be a critical region of size 'a' such that $\omega = \left\{x \in S : \frac{f(x,\theta_1)}{f(x,\theta_0)} > k\right\} \Rightarrow \omega = \left\{x \in S : \frac{L_1}{L_0} > k\right\}$ and $\overline{\omega} = \left\{x \in S : \frac{L_1}{L_0} < k\right\}$, where L_0 and L_1 are the likelihood functions of the sample observations $x = (x_1, x_2, ..., x_n)$, under H_0 and H_1 respectively. Then prove that w is the most powerful critical region of the test $H_0 : \theta = \theta_0$ against the alternative $H_1 : \theta = \theta_1$. (20)

6. (a) Discuss about method of maximum likelihood estimators. (10)

(b) Obtain $100(1-\alpha)\%$ confidence intervals for the parameters

(i) θ and

(ii) σ^2 of the normal distribution

$$f(x, \theta : \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{\frac{-1}{2}\left(\frac{x-\theta}{\sigma}\right)^2\right\}, -\infty < x < \infty . \quad (10)$$

7. (a) Show that for the distribution $dF(x) = \theta e^{-x\theta}$, $0 < x < \infty$, central confidence limits for large samples with 95% confidence coefficients are given by $\theta = \left(1 \pm \frac{1.96}{\sqrt{n}}\right) / \sqrt{x}$. (10)

(b) Obtain the most powerful test for testing the mean $\mu = \mu_0$ against $\mu = \mu_1, (\mu_1 > \mu_0)$, when $\sigma^2 = 1$ in a normal population. (10)

8. (a) The following table shows the lives in hours of four batches of electric lamps. Batches

(i) 1600, 1610, 1650, 1680, 1700, 1720, 1800

(ii) 1580, 1640, 1640, 1700, 1750

(iii) 1460, 1550, 1600, 1620, 1640, 1660, 1740, 1820

(iv) 1510, 1520, 1530, 1570, 1600, 1680.

Perform an analysis of variances of these data and show that a significance test does not reject their homogeneity. (10)

(b) Describe the technique of the analysis of variance. Write down the ANOVA table for two way layout. (10)

DE–8339 **21**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) State and prove Luca's Theorem. (10)

   (b) Prove that cross ratio is invariant under a linear transformation. (10)

2. (a) Prove : A bilinear transformation $W = \frac{az+b}{cz+d}$, where $ad - bc \neq 0$ maps the real axis into itself if and only if a, b, c, d are real. Further this transformation maps the upper half plane Im $z \geq 0$ into the upper half plane $Im\, W \geq 0$ if and only if $ad - bc > 0$. (10)

   (b) Discuss the transformation $W = \frac{1}{2}\left(z + \frac{1}{z}\right)$. (10)

3. (a) Prove : Let $f(z)$ be a analytic on the set R' obtained from a rectangle R by omitting a finite number of interior points ζ_i. If it is true that $\lim_{z \to \zeta j} (z - \zeta_j) f(z) = 0$, for all j, then $\int_{\partial R} f(z) dz = 0$. (10)

   (b) Prove : If the piecewise differentiable closed curve γ does not pass through the point α, then the value of the integral $\int_{\gamma} \frac{dz}{z - a}$ is a multiple of $2\pi i$. (10)

4. (a) Suppose that $\varphi(\zeta)$ is continuous on the arc γ, then the function $F_n(z) = \int_\gamma \frac{\varphi(\zeta)d\zeta}{(\zeta - z)^n}$ is analytic in each of the regions determined by, and its derivative $F'_n(z) = n\,F_{n+1}(z)$. (10)

(b) State and prove Schwarz's Theorem. (10)

5. (a) State and prove Laurent series. (10)

(b) Prove : Let f be analytic in $0 < |z - a| < \delta$, then $\text{Res}_{z=a} f(z)$ is the unique complex number A for which $f(z) - A(z-a)^{-1}$ has a primitive root in $0 < |z - a| < \delta$. (10)

6. (a) Derive the canonical product representation for $\sin \pi z$. (10)

(b) State and prove Hadamard's theorem. (10)

7. (a) Prove : There exists a basis $\{\omega_1, \omega_2\}$ such that the ratio $\tau = \omega_2 / \omega_1$ satisfies the following conditions :

(i) $\text{Im}_\tau > 0$

(ii) $-\frac{1}{2} < \text{Re}_\tau \leqq \frac{1}{2}$

(iii) $|\tau| \geqq 1$

(iv) $\text{Re}_\tau \geqq 0$ if $|\tau| = 1$. The ratio τ is uniquely determined by these conditions, and there is a choice of two, four or six corresponding basis. (10)

(b) Prove : The zeros $a_1, \cdots, a_n$ and poles $b_1, \cdots, b_n$ of an elliptic function satisfy $a_1, \cdots, a_n \equiv b_1, \cdots, b_n \pmod{M}$. (10)

8. (a) Discuss about differential equation $W = \wp(z)$. (10)

(b) Prove : Every point τ in the upper half plane is equivalent under the Congruence subgroup mod 2 to exactly one point in $\overline{\Omega} U \Omega'$. (10)

DE–8340 **22**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

TOPOLOGY AND FUNCTIONAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Let X be a topological space and A an arbitrary subset of X. Then prove that $\overline{A} = \{x :$ each neighbourhood of x intersects A$\}$.

   (b) Let X be a second countable space. Then prove that any open base for X has a countable subclass which is also an open base.

2. (a) Prove that a topological space is compact if every basic open cover has a finite subcover.

   (b) State and prove Tychonoff's theorem.

3. (a) Prove that the product of any nonempty class of Hausdorff spaces is a Hausdorff space.

   (b) Prove that every compact subspace of a Hausdorff space is closed.

4. (a) Prove that the product of any nonempty class of connected spaces is connected.

   (b) Let X be a compact Hausdorff space. Then prove that X is totally disconnected if and only if it has an open base whose sets are also closed.

5. (a) If N is a normed linear space and x_0 is a non zero vector in N, then prove that there exists a functional f_0 in N^* such that $f_0(x_0) = \|x_0\|$ and $\|f_0\| = 1$.

(b) State and prove open mapping theorem.

6. (a) If M and N are closed linear subspace of a Hilbert space H such that $M \perp N$, then prove that the linear subspace $M + N$ is also closed.

(b) If M is a closed linear subspace of a Hilbert space H then prove that $H = M \oplus M^{\perp}$.

7. (a) Prove that a closed convex subset C of a Hilbert space H contains a unique vector of smallest norm.

(b) Let H be a Hilbert space and let f be an arbitary functional in H^*. Then prove that there exists a unique vector y in H such that $f(x) = (x, y)$ for every x in H.

8. (a) If A_1 and A_2 are self-adjoint operators on H, then prove that their product $A_1 A_2$ is self adjoint iff $A_1 A_2 = A_2 A_1$.

(b) If P is a projection on H with range M and null space N, then prove that $M \perp N$ iff P is self adjoint and in this case $N = M^{\perp}$.

DE–8341 **23**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

GRAPH THEORY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Define path. Prove that in a graph G, any $u - v$ walk contains a $u - v$ path.

   (b) Let G be a (p, q) connected graph. Then prove that $q \geq p - 1$.

   (c) Let T be a spanning tree of a connected graph G. Prove.

2. (a) State and prove Cayley's formula.

   (b) Prove that a graph is bipartite if and only if it contains no odd cycle.

3. (a) State and prove Menger's theorem.

   (b) Define Hamiltonian and prove that every Hamiltonian graph is 2-connected.

4. (a) State and prove Chavatal theorem.

   (b) Show that the connectivity $K(G)$ of $G \leq$ the edge connectivity $K'(G)$ of $G \leq \delta$.

5. (a) Prove that for any graph G with 6 points, G or $\overline{G}$ contains a triangle.

   (b) Prove that $\alpha + \beta = p$.

   (c) Prove that $\gamma(m, n) = \gamma(n, m)$.

6. (a) For each positive integer k, show that there exists a graph with girth atleast k and chromatic number atleast k.

   (b) Let $D = (V, A)$ be a digraph each of whose induced subdigraphs has a Kernel. For $v \in V$, let $L(v)$ be an arbitrary list of atleast $d^+(v) + 1$ colours. Then prove that D admits an L-colouring.

7. (a) Let G be a connected plane graph, and let e be an edge of G that is not a cut edge. Then prove that $(G \setminus e)^* \cong G^* / e^*$.

   (b) State and prove Tait's theorem.

8. (a) If two digraphs are isomorphic then corresponding points have the same degree pair.

   (b) Prove that each vertex of a disconnected tournament D with atleast p points $(p \geq 3)$ is contained in a directed cycle of length k, for every k, $3 \leq k \leq p$.

DE–8342 **24**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

PROGRAMMING IN C/C++

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries equal marks.

$(5 \times 20 = 100)$

1. (a) Explain '*compilers*' and '*interpreters*' with atleast two differences. (10)

   (b) Explain library functions used in C language in details with examples. (10)

2. (a) Explain the general structure of a C Program. (10)

   (b) Explain constants and variables and their declaration with an example. (10)

3. (a) Explain puts( ) and putchar( ) functions with examples. (10)

   (b) Write a C++ Program to generate Fibonacci series. (10)

4. (a) Explain the order of precedence of 'arithmetic operators' with an example. (10)

   (b) Write short notes on:-

      (i) Counters

      (ii) increment and

      (iii) decrement operators with examples. (10)

5. (a) Explain about '*Relational operators*' in details with examples. (10)

   (b) Explain in details for loops and its variants with examples. (10)

6. (a) Define Arrays and Initialization of arrays in details with examples. (10)

   (b) Write a C program to multiply two 3×3 matrices. (10)

7. (a) Define structures, pointers and functions with examples. (10)

   (b) Write a C program to print the given five digit number in reverse order. (10)

8. (a) Explain the commands used to open and close the file? Explain in details with example. (10)

   (b) Explain formatted input and output with examples. (10)

DE–8343 **25**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

DISCRETE AND COMBINATORIAL MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) How many words can be formed from the letters of the word 'MISSISSIPPI'? (10)

   (b) Determine the coefficient of x^8 in $\frac{1}{(x-3)(x-2)^2}$. (10)

2. (a) Using a Ferrers graph, show that the number of partitions of n is equal to the number of partitions of $2n$ into n summands. (10)

   (b) Solve the recurrence relation $a_n - 3a_{n-1} = n,\ n \geq 1$, $a_0 = 1$. (10)

3. (a) In how many ways can 3 member team persons be selected out of 15 persons so as to include always one particular person in the team always? (10)

   (b) In how many ways can four $w's$, four $x's$, four $y's$, and four $z's$ be arranged so that there is no consecutive quadrupule of the same letter? (10)

4. (a) Determine the number of positive integers n, where $1 \leq n \leq 100$ and n is not divisible by 2, 3 or 5. (10)

   (b) Define 'derangements' and find the number of derangements of 1, 2, 3 and 4. (10)

5. (a) Let $A = \{1,2,3,4\}$, $B = \{u,v,w,x,y,z\}$. How many one-to-one functions $f : A \to B$ satisfy none of the following conditions :

$C_1; f(1) = u$ or v , $C_2 = f(2) = w$, $C_3 : f(3) = w \; or \; x$, $C_4 : f(4) = x, y$ or z . (10)

(b) Explain Polya's fundamental theorem. (10)

6. (a) Consider for finite sets A, B, where $|A| = m \geq n = |B|$, let

$A = \{a_1, a_2, ..., a_m\}, B = \{b_1, b_2, .., b_n\}$, and $S = \{f : A \to B\}$. Show that $|S| = n^m$. (10)

(b) Explain Polya's generalization theorem. (10)

7. (a) Define 'modular' and 'distributive' lattices with an example. (10)

(b) Prove : Given a finite Boolean algebra $\mathfrak{B}$ with atoms $x_1, x_2, ..., x_n$, each $x \in \mathfrak{B}$ $x \neq 0$, can be written as a sum of atoms uniquely, up to order. (10)

8. (a) Write the Boolean expressions in an equivalent sum of products and product of sums canonical form in X_1, X_2 and X_3.

(i) $X_1 \wedge X_2$

(ii) $X_1 \vee X_2$

(iii) $(X_1 \vee X_2) \wedge X_3$. (10)

(b) Explain : Karnaugh maps with an example. (10)